AFC–10695

Sub. Code
1BIT6C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Information Technology

SOFTWARE ENGINEERING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Software Engineering.

2. List the factors that influence Quality and Productivity.

3. Name any two software cost estimation techniques.

4. What is waterfall life cycle?

5. What are the umbrella activities for a software process?

6. State the merits of incremental model.

7. Write the drawbacks of spiral model.

8. List the process maturity levels of CMM.

9. What is the difference between the "Known Risks" and "Predictable Risks"?

10. Define Software Quality.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe the advantages and disadvantages of size measures.

Or

(b) Explain the activities in software development.

12. (a) Write a short note on software cost estimation techniques.

Or

(b) Explain in detail about Functional Modeling.

13. (a) Illustrate the design concepts.

Or

(b) Explicate the design steps of the transform mapping.

14. (a) What is Black box testing? Explain.

Or

(b) List and explain the various testing strategies for conventional software.

15. (a) Illustrate the quality concepts in detail.

Or

(b) Discuss the activities of SQA.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Draw and Explain the Phase Life Cycle Model.

17. Illustrate the steps involved in the prototyping.

18. Explicate the different phases involved in software implementation.

19. Discuss the following testing mechanisms.

    (a) Stress testing

    (b) White box.

20. Give a brief account on formal technical reviews.

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AFC–10691

Sub. Code
1BIT3C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Third Semester

Information Technology

PROGRAMMING IN C++ AND ALGORITHMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Mention the use of Object Oriented Programming?

2. What are the advantages of function prototypes in C++?

3. Define class.

4. What is the benefit of scope resolution operator in C++?

5. When do you use the protected visibility specifier to a class member?

6. Define virtual base class.

7. What do you mean by topological sorting?

8. Write the condition for Binary Search tree.

9. What is greedy algorithm?

10. How do you find the Fibonacci number?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the concepts of functions in C++.

Or

(b) Discuss the various operators available in C++.

12. (a) Differentiate call by value and call by reference with an example.

Or

(b) Explain Constructors and Destructors in C++ with a program.

13. (a) Explicate Type conversions in C++ with a program.

Or

(b) Illustrate the concepts of Polymorphism.

14. (a) What is Backtracking? Explain with an example.

Or

(b) What is the lower bound of the sorting problem? Discuss it.

15. (a) What is Huffman codes? Illustrate them.

Or

(b) Discuss the Dijkstra's Algorithm.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What is virtual function? Explain with a sample program.

17. Discuss the following :

    (a) Copy Constructor

    (b) Dynamic Constructor

18. How arithmetic operators are overloaded using friend? Write a program to demonstrate it.

19. Explain Strassen's matrix product algorithm.

20. Discuss Floyd and Warshall Algorithms.

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AFC–10692

Sub. Code
1BIT4C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fourth Semester

Information Technology

JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write any four features of Java.

2. What is the difference between the >> and >>> operators?

3. How can you declare an arrays?

4. Specify the use of 'this' operator.

5. Define abstract class.

6. Which package is always imported by default?

7. List out the uses of Java API packages.

8. Name any two container classes.

9. What do you mean by exceptions?

10. How can a dead thread be restarted?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe the structure of a JAVA program.

Or

(b) List and explain the operators available in Java with examples.

12. (a) Elaborate the concepts of method overriding with an example.

Or

(b) Write a Java program to find the smallest of three numbers using constructor.

13. (a) What is wrapper class? Give an example.

Or

(b) Write a Java program to arrange n numbers in ascending order.

14. (a) Explain the threading exceptions.

Or

(b) Discuss the various types of errors with an example.

15. (a) How can you design a web page using applet?

Or

(b) Write a short note on graphics class in Java.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write brief note on WWW and web browsers.

17. Write a Java program to find biggest of three numbers using IF statement.

18. Define interfaces. Write a Java program to implement the hierarchical inheritance.

19. Write a Java applet program to design the student registration form.

20. How applets differ from applications? Describe it.

AFC–10693

Sub. Code
1BIT5C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Information Technology

DATABASE MANAGEMENT SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is database system?

2. Write any two E-R design issues.

3. What is normalization?

4. What is functional dependency?

5. Define transactions.

6. State the benefits of distributed transactions.

7. What is Data Integrity?

8. Define View.

9. What is Stored procedure?

10. State the uses of Packages.

Part B $(5 \times 5 = 25)$

Answer **all** the questions choosing either (a) or (b).

11. (a) Describe the semi structured databases in detail.

Or

(b) What is E-R model? Explain its components of it.

12. (a) Write a short note on First Normal form.

Or

(b) Illustrate about the functional dependency theory.

13. (a) Elucidate the Server system architecture in detail.

Or

(b) Briefly discuss about Distributed systems.

14. (a) Explain the procedures to create Indexes with an example.

Or

(b) Illustrate the steps involved in creating Synonyms with an example.

15. (a) How do you create the implicit cursor in SQL? Explain with an example.

Or

(b) Write the salient features of PL/SQL.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Briefly explain the transaction management.

17. Discuss the 4NF with an example.

18. Give a brief account on parallel databases.

19. Briefly explain the concepts of Data Integrity.

20. Write PL/SQL program to find smallest of three numbers.

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AFC–10694

Sub. Code
1BITE2A

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Information Technology

Elective – COMPUTER NETWORKS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is networking?
2. Expand ATM.
3. What is the expansion of SLIP?
4. What is Error Correction?
5. Define Cell.
6. What is ICMP?
7. List out any two Functions of Transport Layer.
8. What is Crash Recovery?
9. Define Buffering.
10. Expand JPEG.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write short notes on Network Standardization.

Or

(b) Briefly explain about ISDN.

12. (a) Specify in brief about PPP protocol.

Or

(b) Briefly explain about Multimedia Access protocol.

13. (a) Write notes on :

(i) Firewall

(ii) RARP.

Or

(b) Discuss about any one Congestion Control algorithm.

14. (a) Write notes on :

(i) Flow Control

(ii) UDP.

Or

(b) Discuss about Transport Services.

15. (a) Briefly explain about DNS.

Or

(b) Write short notes on Data Compression.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss about Communication Satellites.

17. Explain Carrier Sense Multiple Access Protocols.

18. Illustrate designing Issues in Network Layer.

19. Specify Elements of Transport Protocols.

20. Discuss about Network Security.

AFC–10696

Sub. Code
1BIT6C3

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Information Technology

WEB DESIGN TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What does <tr> tag stand for?

2. Mention any two special characters in HTML.

3. Give the syntax for 'while' statement in JavaScript.

4. List any two arithmetic operators in JavaScript.

5. What is an Iteration?

6. Mention the use of a Subscripted Array.

7. What is an event?

8. Give examples for mouse based events.

9. What is embedding?

10. Give an example for BGSOUND element.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Give a short note on Formatting Text in HTML with examples.

Or

(b) Create a list of friends and places to visit using nested lists.

12. (a) Write a JavaScript program to calculate the biggest of three numbers.

Or

(b) Write a short note on Increment and Decrement operators with examples.

13. (a) What are the scope rules in JavaScript? Explain.

Or

(b) Briefly explain about Boolean and Number objects.

14. (a) Give a short account on Advanced filters in DHTML.

Or

(b) Discuss about flip filters in DHTML.

15. (a) Elucidate about Simple data binding.

Or

(b) Illustrate the IMG element with examples

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Create a mark list in HTML for 10 students using tables and format them accordingly.

17. List and explain the control structures in JavaScript.

18. Explain the different types of Arrays in JavaScript with examples.

19. Explain and write examples for the following using DHTML:

    (a) Making Text glow

    (b) Motion with blur

    (c) Image masks.

20. Write in detail about Animation and Path control.

AFC–10729

Sub. Code
1BITSA1/ 1BIT1A1

U.G. DEGREE EXAMINATION, APRIL 2018

Information Technology

Allied : DISCRETE MATHEMATICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is meant by Compound statement?

2. Define Tautology.

3. Write the inference rules for Modus Ponens and Modus Tollens.

4. Define Principal Disjunctive Normal Form.

5. Define degree of a vertex.

6. What is a Circuit?

7. Define Tree.

8. What is meant by Minimum Spanning Tree?

9. Define Binary relation from one set to another.

10. What is Partial ordering? Given an example.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Construct the truth table for $(P \to Q) \wedge (Q \to P)$.

Or

(b) Show the following implications:

(i) $(P \wedge Q) \to (P \to Q)$

(ii) $P \to (Q \to P)$.

12. (a) Determine whether the conclusion C follows logically from the premises H1 and H2.

(i) H1 : P → Q H2:7 (P ∧ Q) C:7P

(ii) H1 : P → Q H2 : Q C : P .

Or

(b) Prove that P, P → Q, Q → R ==> R .

13. (a) Explain Isomorphism of two graphs with an example.

Or

(b) Explain Bipartite Graph with an example.

14. (a) Explain Prim's Algorithm to find minimum spanning Tree of a Graph.

Or

(b) Write short notes on Hamiltonian Graph.

15. (a) Write the properties of Lattice.

Or

(b) Define Complement of an element of a lattice and Complement Lattice.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain Conditional and Bi-Conditional statements with truth tables.

17. Obtain the Principal Conjunctive Normal Forms of $(7P \rightarrow R) \wedge (Q \leftrightarrow P)$.

18. Show that the number of vertices of odd degree in a graph is always even.

19. Explain Dijkstra's Algorithm to find shortest path between a specified vertex to another Specified vertex.

20. State and prove the isotonic property of a lattice.

AFC–10730

Sub. Code
1BITSA2

U.G. DEGREE EXAMINATION, APRIL 2018

Information Technology

Allied — OPERATION RESEARCH

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is an OR Model?

2. What are the application areas of OR?

3. Define Non-degenerate basic solution.

4. What is a Slack variable?

5. What is the dual optimality test?

6. What is an Integer Programming Problem?

7. Give four application areas of Assignment Problem.

8. What is the basic assumption for solving an Assignment Problem?

9. Write the commonly used methods of finding BSF.

10. What is an unbalanced Transportation Problem?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write about development of OR.

Or

(b) What are the main components of an OR Model?

12. (a) Explain the various steps involved in the Simplex Method.

Or

(b) Use the graphical method to solve the following LPP:

Maximize $Z = 10x_1 + 15x_2$

Subject to

$$2x_1 + x_2 \leq 26$$
$$2x_1 + 4x_2 \leq 56$$
$$-x_1 + x_2 \leq 5$$

and $x_1, x_2 \geq 0$.

13. (a) Solve the following LPP by Dual Simplex Method

Minimize $Z = 3x_1 + 2x_2$

Subject to

$$7x_1 + 2x_2 \geq 30$$
$$5x_1 + 4x_2 \geq 20$$
$$2x_1 + 8x_2 \geq 16$$

and $x_1, x_2 \geq 0$.

Or

(b) Explain the steps involved in Branch and Bound Method.

14. (a) What are the job-assignments which will minimize the cost for the following?

Machines

		W	X	Y	Z
	A	18	24	28	32
Jobs	B	8	13	17	18
	C	10	15	19	22

Or

(b) Write short notes on Travelling Salesman Problem.

15. (a) Solve the following by Least Cost Method.

To

		D	E	F	Supply
	A	6	4	1	50
From	B	3	8	7	40
	C	4	4	2	60
	Demand	20	95	35	

Or

(b) Solve the Transportation Problem by VAM.

Destination

		A	B	C	D	Supply
	1	11	20	7	8	50
Source	2	21	16	20	12	40
	3	8	12	18	9	70
	Demand	30	25	35	40	

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the scope of OR.

17. Solve the following LPP by Simplex Method:

Maximize $Z = 3x_1 + 2x_2$

Subject to

$$x_1 + x_2 \le 4$$
$$x_1 - x_2 \le 2$$

and $x_1, x_2 \ge 0$.

18. Solve the following LPP using duality

Minimize $Z = 8x_1 + 10x_2$

Subject to

$$8x_1 + 4x_2 \ge 150$$
$$3x_1 + 9x_2 \ge 100$$

and $x_1, x_2 \ge 0$.

19. Solve the following travelling salesman Problem:

	1	2	3	4	5
1	–	6	12	6	4
2	6	–	10	5	4
3	8	7	–	11	3
4	5	4	11	–	5
5	5	2	7	8	–

20. Find the optimal solution of the following transportation problem using MODI Method.

		Destination				
		D1	D2	D3	D4	Supply
Origin	O1	6	1	9	3	70
	O2	11	5	2	8	55
	O3	10	12	4	7	70
	Demand	85	35	50	45	

AFC–10758

Sub. Code
1BITSA3

U.G. DEGREE EXAMINATION, APRIL 2018

Information Technology

Allied — ACCOUNTING PRINCIPLES AND COMPUTER APPLICATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

Answer in One or Two sentences:

1. What is accounting? Write its main functions.

2. Write the difference between Final Accounts with Adjustment and without Adjustment.

3. List out the advantages of Trial Balance.

4. Capital + Liabilities = ————————

5. What are the steps involved in Accounting Cycles?

6. Define "Balance Sheet".

7. What is the difference between sales and sales return book?

8. List the different types of Subsidiary Books.

9. Define the term "Favourable Balances".

10. What is called Bank Reconciliation Statement?

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) From the following balances extracted from the books of a Trader the close of the accounting year ending 31st December 2000, prepare Profit and Loss Account.

Trial Balance as on 31st December 2000

Particular	Rs.	Particular	Rs.
Gross Profit	1,53,000	Apprenticeship Premium	5,000
Salaries	46,500	Dividend Received	12,000
Rent (Office)	12,300	Interest Received	8,000
Fire Insurance	2,700	Rent Received	5,000
Premium	1,500		
Discount Allowed	7,500		
Carriage Outward	6,300		
Bad Debts	750		
Printing and stationary	21,500		
Selling expenses	1,050		
Rent, Rates and Taxes	8,000		
Loss by Fire (not covered by Insurance)	2,000		
Legal Charges			
Goodown Rent	12,700		
Depreciation on Office			
Equipment	5,500		
Repairs and Maintenance	4,800		
Bank Charges	1,200		

Salaries due but not paid Rs. 2,500. Fire Insurance Premium has been in advance to the extent of Rs. 300. Rent, rates and Tax outstanding Rs. 150.

Or

(b) Explain the Accounting Concepts and Conventions.

12. (a) Difference between Journal and Ledger.

Or

(b) Journalise the following transactions and post them to the ledger accounts concerned:

1991		Rs.
January 1	Purchased goods for cash	2,000
January 3	Sold goods to Karim	500
January 10	Received from Karim	500
January 15	Purchased machinery for cash	1,000
January 20	Cash sales	300
January 25	Sold goods to Rahim and Sons	600
January 28	Received from Rahim and Sons	590
	Discount allowed	10
January 30	Paid Rent	50
January 31	Paid Salaries	100

13. (a) Prepare a petty cash book on the imprest system from the following:

2005		Rs.
March 1	Received	1000 for petty cash
March 2	Paid bus fare	10
March 3	Paid Cartage	25
March 4	Paid for postage and telegram	50
March 5	Paid for stationary	40
March 6	Paid for postage and telegram	50
March 7	Paid for sundry exp.	40

Or

(b) Enter the following transactions in simple Cash Book.

2006		Rs.
January 1	Cash in hand	12,000
January 5	Received from Ramesh	3,000
January 7	Paid Rent	3,000
January 8	Sold goods	7,000
January 10	Paid Sohan	2,000

14. (a) From the following balances extracted from the books of X and Co., prepare a trading and profit and loss account and balance sheet on 31st December, 1991.

	Rs.		Rs.
Stock on 1st January	11,000	Returns outwards	500
Bills receivables	4,500	Trade expenses	200
Purchases	39,000	Office fixtures	1,000
Wages	2,800	Cash in hand	500
Insurance	700	Cash at bank	4,750
Sundry debtors	30,000	Tent and taxes	1,100
Carriage inwards	800	Carriage outwards	1,450
Commission (Dr.)	800	Sales	60,000
Interest on capital	700	Bills payable	3,000
Stationary	450	Creditors	19,650
Returns inwards	1,300	Capital	17,900

The stock on 21st December, 1991 was valued at $25,000.

Or

(b) From the following information of Johnson's Ltd. on 31st March, 2003 you are required to prepare the Trading, Profit and Loss A/c

	Rs.		Rs.
Opening Stock	5,000	Capital	89,500
Bills Receivable	22,500	Commission (Cr.)	2,000
Purchases	1,95,000	Return Outward	2,500
Wages	14,000	Trade Expenses	1,000
Insurance	5,500	Office Fixtures	5,000
Sundry debtors	1,50,000	Cash in Hand	2,500
Carriage inwards	4,000	Cash at Bank	23,750
Commission (Dr.)	4,000	Rent & Rates	5,500
Interest on capital	3,500	Carriage Outward	7,250
Stationary	2,250	Sales	2,50,000
Returns inwards	6,500	Bills Payable	15,000
Creditors	98,250		
Closing Stock	12,500		

15. (a) Explain the causes for the differences in two balance.

Or

(b) From the following particulars of M/s Ananaya industries, prepare bank reconciliation statement as on December 31, 2006.

(i) Bank balance as per cash book Rs. 32,500

(ii) Cheques deposited into bank but not credited upto December 31, 2006 Rs. 8,900

(iii) Cheques issued but not presented for payment Rs. 12,500

(iv) Bank credited Rs. 5,000 for receiving dividend through Electronic Clearing System.

(v) Bank charges debited by Bank Rs. 400.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Mr. Ramu has the following transactions in the month of July. Record them into the journal and show postings in the ledger and balance the accounts.

July		Rs.
1st :	Ramu started business with a capital of	75,000
1st :	Purchased goods from Manu on credit	25,000
2nd :	Sold goods to Sonu	20,000
3rd :	Purchased goods from Meenu for cash	15,000
4th :	Sold goods to Tanu for cash	16,000
5th :	Goods retuned to Manu	2,000
6th :	Bought furniture for	15,000
7th :	Bought goods from Zenu	12,000
8th :	Cash paid to Manu	10,000
9th :	Sold goods to Jane	13,500
10th :	Goods returned from Sonu	3,000
11th :	Cash received from Jane	5,500
12th :	Goods taken by Ramu for domestic use	3,000
13th :	Returned Goods to Zenu	1,000
14th :	Cash received from Sonu	12,000
15th :	Bought machinery for	18,000
16th :	Sold part of the furniture for	1,000
17th :	Cash paid for the purchase of bicycle for Ramu's son	1,500
19th :	Cash sales	15,000
20th :	Cash purchases	13,500

17. A Government Organization receives Grant in Aid amounting to Rs. 10,00,000 on 1.4.2007. Against this grant the organization spends the following amount during the period ended on 3 1.03.2008.

Sl. No	Date	Particulars	Rs.
1.	01.04.2007	Opened Bank Account by depositing	
		Cheque of Grant in Aid	10,00,000
2.	05.04.2007	Withdrawn Cash	60,000
3.	06.04.2007	Purchase of Table and Chairs	
		From Indian Store	1,05,000
4.	06.04.2007	Purchase of Computer From	
		M/s Excel enterprises	1,50,000
5.	10.04.2007	Cash purchases of 2 ceiling fan	8,000
6.	10.05.2007	Travelling Advance to A	5,000
7.	11.06.2007	Printing Expenses	2,000
8.	12.06.2007	A submit the claim for expenses of	2,000
9.	15.07.2007	Medical Advance to B by Cheque	10,000
10.	24.07.2007	A submit the claim for expenses of	1,000
11.	30.07.2007	Salaries paid by bank	15,000
12.	15.08.2007	A submit the claim for expenses of	1,000
13.	25.08.2007	B, Submitted the Medical Expenses and	
		Balance Amount returned to office	5,000
14.	30.08.2007	Salary paid by bank	15,000

Sl. No	Date	Particulars	Rs.
15.	30.09.2007	Salary Paid by Bank	15,000
16.	12.10.2007	Loan Given to C by cheque	12,000
17.	30.10.2007	Salary paid by bank	15,000
18.	1.11.2007	A submit the claim for expenses of	1,000
19.	05.12.2007	Further Advance to A	10,000
20.	25.12.2007	Cash Withdrawn	15,000
21.	30.12.2007	Salary paid by Cash	15,000
22.	15.01.2008	Loan Received from C	12,000
23.	25.01.2008	Printing expenses	5,000
24.	30.01.2008	Salary paid by bank	15,000
25.	15.02.2008	Salary paid by bank	13,500
	15.02.2008	Salary accrued but not yet paid to D	1,500
26.	15.02.2008	Miscellaneous Expenses	12,000
27.	17.02.2008	Salary Paid to 'D" in Cash	1,500
28.	27.02.2008	Salary paid by Bank	15,000
29.	05.03.2008	Printing expenses	3,000
30.	11.03.2008	Telephone expenses by cheque	15,000
31.	15.03.2008	Advance to Mr. Verma (Employee)	8,000
32.	31.03.2008	Payment to India stores by cheque	50,000
33.	31.03.2008	Payment to M/s Excel enterprises	72,500

Prepare trial balance, income and expenditure account and balance sheet as on 31.03.2008

18. Rakesh commenced business on 1.1.05 with Rs. 15,000 cash. Prepare two column cash book taking into consideration the following:

2005

January		Rs.
3	Paid cash to Suresh	800
	Allowed discount	10
5	Received from Mohan	900
	Allowed him discount	15
7	Purchased goods for cash	6,000
10	Sold goods for cash	4,000
12	Purchased goods for cash	3,000
15	Sold goods for cash	2,000
17.	Amount of Rs. 600 was payable to Suresh which has been paid after deducting 2% discount.	
20.	Goods Purchased for cash	4,410
25.	Deposited in bank	500
28.	Received from Dinesh	2,000
	Allowed him discount	30
29.	Paid Rs. 340 to Mahesh in full settlement of his account of	400
29.	Withdrew for personal use	150
30.	Purchased goods for cash	9,600
31.	Purchased goods from Dinesh for cash	588
31.	Sold goods to Mohan for cash	2,850

19. From the following Trial Balance of Gurdeep Singh as at 31st December, 2007, prepare Trading and Profit and Loss Account and Balance Sheet:

Dr. Balance	R.s	Cr. Balances	Rs.
Opening Stock	15,500	Capital	60,000
Land and Building	35,000	Loan from Mrs. Gurdeep Singh	30,000
Machinery	50,000	@%	9,600
Furniture and Fixtures	5,000	Sundry creditors	2,100
Purchases	1,06,000	Purchases returns	2,07,300
Salaries	11,000	Sales	1,200
General Expenses	2,500	Discount	
Rent	3,000		
Postage and Telegrams	1,400		
Stationery	1,300		
Wages	26,000		
Freight on Purchases	2,800		
Carriage on Sales	4,000		
Repairs	4,500		
Sundry Debtors	30,000		
Bad Debts	600		
Cash in Hand	100		
Cash at Bank	6,400		
Sales Returns	5,100		
	3,10,200		3,10,200

The following further information was given;

(a) Wages for December, 2007 amounting to Rs. 2,100 have not yet been paid.

(b) Included in General Expenses is Insurance Premium Rs. 600, paid for the year ending 31st March, 2008.

(c) A provision for doubtful debts @ 5% on debtors is necessary.

(d) Depreciation is to be charged as follows:

Land and Building 2%, Machinery 10% and Furniture and Fixtures 15%.

(e) The loan from Mrs. Gurdeep Singh was taken on 1st July, 2007. Interest has not been paid yet.

(f) The value of stock on hand on 31st December, 2007 was Rs. 14,900.

20. (a) From the following particulars, prepare Bank Reconciliation statement as on December 31, 2006.

(i) Balance as per Cash Book Rs.4,200

(ii) Cheques issued but not presented for payment Rs. 2,000

(iii) Cheques deposited but not collected Rs. 3,000

(iv) Bank charges debited by the bank Rs. 250.

(b) From the following particulars of Neha and Co. prepare Bank Reconciliation Statement on March 31, 2006

	Rs.
Overdraft as per pass book	16,500
Interest on overdraft	1,600
Insurance premium paid by the bank	800
Cheques deposited but not yet credited	5,500
Cheques issued but not present for payment	6,000
Wrongly credit to firm account by the bank	1,000

AFC–10813

Sub. Code
IBIT2C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Information Technology

PROGRAMMING IN C AND DATA STRUCTURES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Variables.
2. What is the use of Bitwise operators?
3. Give syntax for Initializing One Dimensional Array.
4. What is Dynamic Memory Allocation?
5. Define Structures.
6. Give syntax for Creating a File.
7. Define Postfix with an example.
8. What is a Queue?
9. What is a Binary Tree?
10. List out any two applications of Trees.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss about Decision Making Statements in C with suitable examples.

Or

(b) Write a C program to find out Sum of series (1+2+3+ .... + n).

12. (a) Specify declaration and initialization of Two Dimensional Array.

Or

(b) Write a C program to find out Total and average using Pointers.

13. (a) Briefly explain about Union.

Or

(b) Write short notes on Unformatted Data File.

14. (a) Briefly discuss about Queue with suitable example.

Or

(b) Illustrate Command Line Parameters.

15. (a) Write short notes on applications of Trees.

Or

(b) Discuss the concept of Lists with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Illustrate Switch ... case statement in C with suitable example.

17. Explain passing Functions to other Function with an example.

18. Specify the concepts of File such as Opening a file, Read/Write a file and Closing a file.

19. Discuss Stack with an example.

20. Exemplify Binary Tree Representation.

AFC–10839

Sub. Code
1BITE3A

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Information Technology

Elective: DISTRIBUTED COMPUTING

(CBCS 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Financial Trading?

2. What are Application Services?

3. Mention the use of Unmarshalling.

4. Define Peer-to-peer networking.

5. Describe a process.

6. Expand RPC.

7. Define Internal Synchronization.

8. How does a DNS get resolved?

9. What is a Transaction?

10. Describe Fault Tolerance.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Mention the responsibilities of the Physical Model of a distributed system.

Or

(b) Give a short account on IP addressing.

12. (a) Explicate about Client–server communication.

Or

(b) How is communication done between distributed objects? Explain.

13. (a) Give a short note on Communication primitives.

Or

(b) What are the requirements of a Distributed File System?

14. (a) How are Physical Clocks synchronized? Clarify.

Or

(b) Write a short note on Distributed mutual exclusion.

15. (a) Interpret the steps to recover from a failed Transaction.

Or

(b) Mention the rule of group communication.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the trends of distributed systems in detail.

17. Elucidate about Request-Reply protocols.

18. Describe in detail about Digital Signing.

19. Discuss about Name Services and the Domain Name System.

20. Enumerate and explain the Fault-tolerant Services.

AFC–10837

Sub. Code
1BIT5C2

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Information Technology

VISUAL PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **ALL** the questions.

1. Write any two advantages of procedures in VB.
2. List out the various data types used in VB.
3. What are the short cut keys used in window forms?
4. What are Active-X controls?
5. What is REF Languages?
6. Write any two uses of Rich Text Box Control.
7. Expand ADODC.
8. How the tables can be connected to VB.
9. What are the uses of Accelerators?
10. What is a message map?

Part B (5 × 5 = 25)

Answer **ALL** the questions, choosing either (a) or (b).

11. (a) How can you declare arrays in VB? Give its syntax.

Or

(b) List out the features of Visual Basic.

12. (a) How can you build a dynamic form at runtime?

Or

(b) Write note on list box and combo box controls with an example.

13. (a) Write a VB program to move a text box using timer control.

Or

(b) Illustrate with an example the common dialog controls.

14. (a) Differentiate ADO and DAO.

Or

(b) Write note on ADODC based control.

15. (a) Write short note on MFC class hierarchy.

Or

(b) Describe about Document/view architecture.

Part C (3 × 10 = 30)

Answer any **THREE** questions.

16. Write a VB program to print the numbers form 1 to 1000 using different loop statements.

17. Write brief note on list box and control box controls with an example.

18. Explain about common dialog controls with an example.

19. Write VB program to read the employee details from table and produce a employee paybill.

20. Explain the various object properties in VC++.

AFC–10838

Sub. Code
1BIT6C2

B.Sc. DEGREE EXAMINATION, APRIL 2018

Sixth Semester

Information Technology

MOBILE COMMUNICATION

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Modulation?

2. Define Wireless transmission.

3. Expand: MAC.

4. What is Routing?

5. What is HIPERLAN?

6. Expand IEEE.

7. What is TCP?

8. Explain snooping.

9. Describe Mobility.

10. Expand HTTP.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is Signal Propagation? Explain in detail.

Or

(b) Write short notes on Antennas.

12. (a) Clarify the concept of SDMA.

Or

(b) Write the a short note on Digital Audio Broadcasting.

13. (a) Clarify the concept of WATM Services.

Or

(b) What are the features of Radio Access Layer?

14. (a) Elucidate the concept of Dynamic host configuration protocol.

Or

(b) How to use selective Retransmission?

15. (a) What are the features of using HTML?

Or

(b) Explain the System Architecture.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the concept of Spread Spectrum.

17. Explain the following

    (a) GSM and DECT

    (b) Localisation and Handover.

18. Explain various features available in Wireless LAN.

19. Describe the Mobile network layer.

20. Explain the WAP Architecture.

AFC–10840

Sub. Code
1BITSA4

U.G. DEGREE EXAMINATION, APRIL 2018

INFORMATION TECHNOLOGY

Allied: E-COMMERCE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define E-Commerce.

2. List the components of I-Way.

3. What is WWW?

4. Write any two advantages of EPS.

5. Specify the features of EDI.

6. Name any two Value Added Networks.

7. What is E-Catalog?

8. How do you filter the information?

9. State the purpose of Digital Copyrights.

10. Define Software Agents.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe the E-Commerce applications.

Or

(b) Write a short note on Internet governance.

12. (a) How to publish Hypertext? Explain with examples.

Or

(b) Illustrate the procedures to design Electronic Payment System.

13. (a) Elaborate the EDI applications available for businesses.

Or

(b) Give a brief account on EDI envelope.

14. (a) What do you mean by Information-based Marketing? Explain its concepts.

Or

(b) What are the tools available for data interface? Explicate any two of them.

15. (a) Why do we need Computer-based education and training? Justify.

Or

(b) Describe the history of software agents.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the components of E-Commerce framework with neat diagram.

17. Explain any two types of Electronic Payment System.

18. Give a brief account on EDI and its Standardization.

19. Illustrate the basic concepts of E-Catalogs.

20. Discuss the characteristics and properties of Software Agents.

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AFC–10858

Sub. Code
1BIT1C1

B.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Information Technology

PRINCIPLES OF INFORMATION TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is connectivity?

2. What are the limitations of using communication system?

3. Write the different types of software.

4. What is Groupware?

5. State the use of Video Conferencing.

6. Expand: ISDN.

7. Define Multimedia.

8. What is Magnetic tape?

9. What is MIS?

10. Mention any two programming languages.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write the evolution of Computer and Communications.

Or

(b) Discuss the various elements of a computer.

12. (a) What is User Interface? Explain its features.

Or

(b) Elucidate the features of Presentation Graphics Software.

13. (a) What do you mean by Voice Communication? Explain in detail.

Or

(b) Clarify the concepts of Modems and Data communication software.

14. (a) Write brief a note on Compression and Decompression techniques.

Or

(b) Describe the Data Management System in detail.

15. (a) What are the steps involved in programming? Explain them.

Or

(b) Elucidate the various generations of computer languages.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss about Ethics of Information Technology.

17. Describe in detail about Application Software.

18. Give a brief note on Shared Resources.

19. Illustrate characteristics of any two storage devices.

20. Explain the following:

    (a) Object Oriented Programming Languages.

    (b) Visual Programming.

AFC–10859

Sub. Code
1BITE1B

B.Sc. DEGREE EXAMINATION, APRIL 2018

Fifth Semester

Information Technology

Elective : MULTIMEDIA TECHNOLOGY AND ITS APPLICATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List out any two Resources for Multimedia Developers.

2. Define Multimedia.

3. List out any two Multimedia Software.

4. What is Hypertext?

5. List out Elements of Text.

6. Define Digital Audio.

7. What is Playback Systems?

8. Define Computer Animation.

9. What are Benefits of Multimedia?

10. Explain Virtual Reality Systems.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about Multimedia Market.

Or

(b) Write short notes on Products and Evaluations.

12. (a) Illustrate Multimedia Computer architecture.

Or

(b) Write notes on Hypertext in Multimedia.

13. (a) Explain characteristics about Audio File Formats.

Or

(b) Briefly explain about Digital Audio in Multimedia.

14. (a) Explain Digital Video in Multimedia.

Or

(b) Briefly discuss characteristics of Digital Video.

15. (a) Discuss about strategy for selecting a tool.

Or

(b) Explain Multimedia Considerations in the Internet

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss Evaluation of Multimedia.

17. Specify Hardware, software and Operating Systems in Multimedia.

18. Illustrate MIDI with real time applications.

19. Discuss about Computer Animation.

20. Explain about Categories of Authoring Tools.